APPROACH THAT I TRIED

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<u>Quick and Dirty Zero Shot approach to get things started otherwise overthing and perfection wont move the stone</u>

I took a look at the input and expeccted output (refereance document shared)and why the classes were rated so high , so we had 2 input files one is video and transcripts, and modelled solution in a way such that

* assume we have a new transcript file what all are the conditions that will make the transcript rating High, low, average
* also from video input , from video and audio what all conditions that will make the video , audio rating higher.

Batch / Video ID	Original Label	Transcript Score (1-5)	Audio Score (1-5)	Video Score (1-5)	Overall Average	Why Transcript Score is Rated	Why Audio Score is Rated	Why Video Score is Rated
B491	Excellent	4	4	3	3.7	Questions asked individually by name and HW was given.	Good interaction tone maintained throughout.	Standard background setup was present.
B472	Excellent	4	3	3	3.3	Good rapport building and asked questions individually.	Tone was adequate for class engagement.	Background and general setup were acceptable.
B552	Good	3	3	4	3.3	Included a solid introduction and assigned HW.	Needed more pleasant verbal interaction.	Branded 8*8 background was properly fixed.
B435	Good	3	3	3	3.0	Good intro and included everyone, but missed HW.	Delivery was balanced with reasonable interaction.	Acceptable camera setup and framing.
B516	Average	3	3	2	2.7	Addressed learners during setup and puzzles; gave HW.	Standard delivery without major vocal issues.	Clarity problems noted with the virtual background.
B435	Average	3	2	2	2.3	Maintained one-to-one queries, but should give HW.	Needed more energetic and pleasant interaction.	Insufficient room lighting affected the image.
B558	Weak	2	2	3	2.3	Lacked good intro/conclusion and joiner introductions.	Needed more energetic and pleasant delivery.	Virtual background was maintained.
B482	Weak	2	2	2	2.0	Skipped rapport for revision; missed joiner intros.	Subdued tone without warm opening engagement.	Camera was turned off while teaching.
B392	Known Quality Issue	3	2	2	2.3	General interactive flow was present.	Needed a happier, warmer communication style.	Virtual background was missing entirely.
B441	Known Quality Issue	2	2	2	2.0	Directly entered class without building rapport.	Needed happier communication with learners.	Tutor's face was not visible on camera.

1.1. Now I have divided the problem into two

One which takes transcripts as input and make the llm as an expert pedagogical auditor evaluating online children's chess classes made the llm behave like an expert pedagogical auditor for evaluating online children's chess classes

TRANSCRIPTS INPUTS : Model scoring is like this
Features

1. asked questions individually / Questions asked individually by name /included everyone /Maitainated one to one queries
2.HW was given / assigned homework / gave homework or missed HW
Good rapport building or Skipped rapport /Directly entered class without building rapport
3.Included a solid introduction /Good intro or Lacked good intro
4.Addressed learners during setup and puzzles
5. General interactive flow was present
6.Lacked conclusion
7. joiner introductions or missed joiner intro

High value feature combination = 4

(Questions asked individually by name + HW was given) or
(Good rapport building + asked questions individually)

Medium value feature combination = 3

(Included a solid introduction + assigned homework) or
(Good intro + included everyone + missed HW) or
(Addressed learners during setup and puzzles + gave homework) or
(Maitainated one to one queries + missed Homeworks) or
(General interactive flow was present)

Low value features combination = 1

(Lacked good intro + lacked conclusion + joiner introductions) or
(Skipped rapport + missed joiner intro) or
(Directly entered class without building rapport.)

Modelling approach for transcripts

The transcript scoring model relied on an expert-guided, rule-grounded NLP approach calibrated against human evaluator rubrics. Speech transcripts were parsed into structured conversational segments—differentiating the opening, interactive puzzle-solving, and lesson closure.

Using few-shot LLM prompting with strict JSON schema constraints, the model extracted discrete pedagogical signals: warm rapport building, individual student callouts by name, continuous check-ins during activities, and homework assignment. Rather than uncalibrated sentiment analysis, these extracted features were evaluated against predefined rubric benchmarks derived from past evaluator feedback, mapping verbal engagement, interactive flow, and pedagogical structure into calibrated quality tiers (Poor, Average, Good, Excellent) with grounded, timestamped justifications.

Here is a well-structured recount of the technical and logistical challenges you faced, written so you can easily include it in your submission write-up, documentation, or interview discussion:

Key Challenges Faced During Development

1. Multimodal API Rate Limits (Gemini):

Google's Gemini models provided the highest quality multimodal evaluations—accurately capturing audio tone, teacher presence, and visual classroom dynamics in synchronized passes. However, strict daily quota ceilings (20 requests/day on the free tier) made iterative prompt engineering, edge-case testing, and automated batch evaluation across multiple recordings unsustainable.

2. Emotion & Tone Degradation with Fast Fallbacks (Groq):

To bypass the request limits and maintain quick iteration speeds, I pivoted to open-weight LLMs hosted on Groq for transcript scoring. While inference speed was exceptional, these text-only models struggled to reliably capture nuanced social-emotional dynamics—such as vocal warmth, genuine pleasantness, and teacher patience—often misinterpreting flat monologues as positive simply based on polite phrasing.

3. Cloud Infrastructure & Credit Blockers (Amazon Bedrock):

I attempted to integrate Amazon Bedrock to access alternative enterprise-grade multimodal foundations (like Claude 3.5 Sonnet). However, account verification delays, credit allocation constraints, and quota restrictions on Bedrock multimodal inference blocked this path within the short 24-hour window.

4. Hardware & Compute Constraints (No Dedicated GPU):

Developing on a local personal machine with only CPU compute meant local model hosting, parameter-efficient fine-tuning (LoRA/PEFT), or running dedicated open-source vision-language models (e.g., LLaVA, Whisper-large, or emotion classifiers) was out of reach. Fine-tuning an open-source classifier on the feedback dataset was not viable locally, forcing the architecture to rely entirely on zero-shot/few-shot cloud API orchestration and careful prompt constraint design.

Areas to improve or research can fix

Due to the time< resource limitations I made a very impatient model, there are many areas to improve like scoring metrics , capture more features from the transcripts like capturing Sentiments, politeness and verbal engagement metrics, pedagogical structure and phase timing metric, adding more conversational balance metrics

1. Conversational Balance & Inclusivity Metrics

* Speaker Talk-Time / Turn-Taking Ratio: The percentage of total words or turns spoken by the tutor versus students. A healthy interactive class benchmarks around 55–65% tutor speech, penalizing passive monologue lecturing (>85% tutor words).

* Student Participation Gini Coefficient / Entropy: Evaluates whether conversation is distributed evenly among all participants or monopolized by 1–2 vocal students while others remain silent.

* Individual Callout Density: Counts the frequency of unique student names mentioned by the tutor, normalized per 15-minute window (e.g., verifying that every enrolled learner is addressed directly by name, especially during puzzle or exercise phases).
* Question-to-Monologue Ratio: The count of interrogative sentences (checking comprehension, inviting answers) relative to purely expository declarative sentences.

Want to research more on the same regarding the metrics and more about capturing features rather than giving LLM zero shot examples.

2. **About video and Audio : How I approached**

Audio and video prediction

class: Excellent , score = 4

Audio : (Good interaction tone maintained throughout) or (Tone was adequate for class engagement)
Video : (Tone was adequate for class engagement.) + (Background and general setup were acceptable.)

class: Good

Audio: (Needed more pleasant verbal interaction.) or (Delivery was balanced with reasonable interaction.)
Video : (Branded 8*8 background was properly fixed.) or (Acceptable camera setup and framing.)

class: Average

Audio : (Standard delivery without major vocal issues.) or (Needed more energetic and pleasant interaction.)
video : (Clarity problems noted with the virtual background.) or (Insufficient room lighting affected the image.)

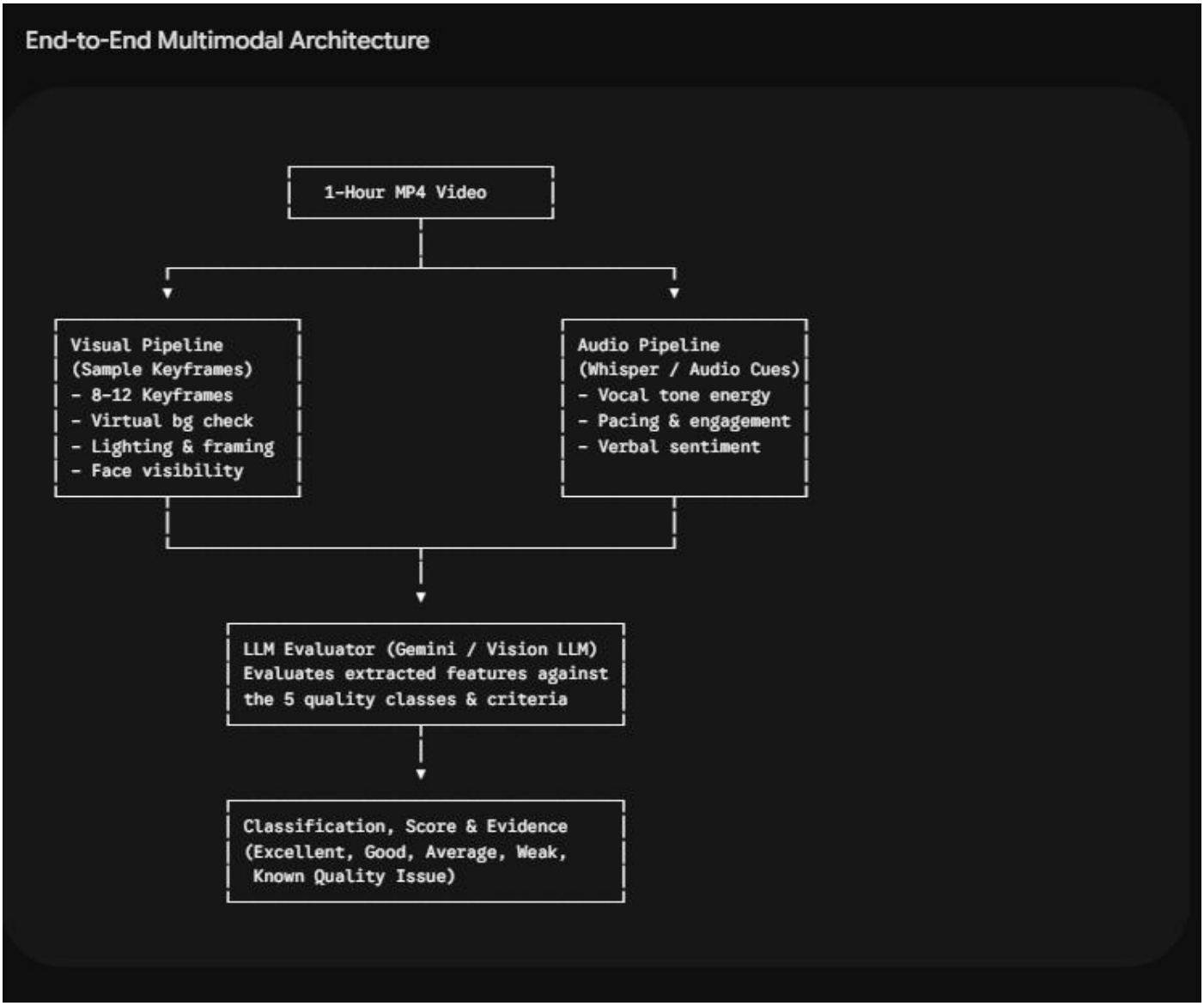
class : weak

Audio : (Needed more energetic and pleasant delivery.) or (Subdued tone without warm opening engagement.)
video : (Virtual background was maintained.) or (Camera was turned off while teaching.)

class : Known Quality Issue

Audio : (Needed a happier, warmer communication style.) or (Needed happier communication with learners.)
video : (Virtual background was missing entirely.) or (Tutor's face was not visible on camera.)

End-to-End Multimodal Architecture



Feature Extraction Pipeline (Frame & Audio Sampling)
Sending 1 hour of raw 30 FPS video (~108,000 frames) is unnecessary and bandwidth-prohibitive. Instead:
Video: Sample 8 to 12 evenly spaced keyframes (e.g., at minutes 2, 5, 15, 30, 45, 55). These capture:
Brand background (8×8 chess branded background vs. missing vs. visual artifacts).
Tutor face visibility and framing.
Room lighting conditions.
Audio: Extract vocal pacing, sentiment cues from transcript intervals, or run audio amplitude/pitch variance checks.

Approach :

To audit 1-hour coaching recordings (.mp4) of up to 500 MB without encountering memory crashes or token context limits, the system uses a decoupled, chunked stream-and-sample pipeline. The incoming video file is written directly to disk in 8 MB chunks rather than loaded into RAM all at once. OpenCV's direct frame-seek pointer (CAP_PROP_POS_FRAMES) then jumps straight to 6–12 evenly distributed timestamps between minutes 2 and 58, skipping intro and outro black screens while bypassing sequential decoding. Each sampled frame is resized to 360p and converted to base64 for fast UI previews, alongside automated pixel analysis to extract brightness and contrast metrics to flag low-light or unlit conditions. This visual telemetry and timeline summary are fed into a structured LLM-as-a-Judge prompt, which evaluates the audio delivery tone and video setup against the five standardized quality tiers (Excellent, Good, Average, Weak, and Known Quality Issue). The prompt strictly checks for critical vetoes—such as the tutor's face not being visible or the virtual background missing entirely—and outputs a validated JSON schema containing the matched rubric rules, timestamps, and observational evidence. Immediately after the audit finishes, the temporary 500 MB disk cache is automatically unlinked and purged, keeping the entire runtime footprint under 100 MB of RAM.

AREAS to improve

1. Better scoring matrix
2. Better capturing models to differentite between students and teachers, sometimes in a frame students background is blur or dark it is affecting the overall consistency of the model regarding the metrics
3. Better video procsessing models, finetuning video specific to the datasets

1. Granular & Multi-Factor Scoring Matrix
Current Limitation: The scoring relies on discrete combinatorial buckets (Scores 1, 3, 4 for transcripts; 0–4 for video) that pool linearly into broad quality tiers (Poor, Average, Good, Excellent). This can mask subtle pedagogical strengths or over-penalize an otherwise high-quality session due to a single missed item.

Proposed Enhancement: Transition from a purely rule-based triage to a weighted multi-criteria decision analysis (MCDA) framework.

Assign normalized continuous weights (0.0–1.0) to atomic features based on pedagogical impact (e.g., student participation and check-ins carry higher weight than puzzle transition latency).

Introduce confidence scoring and soft thresholds alongside hard veto gates. This prevents an outright score drop when an ambiguous edge case occurs (e.g., homework assigned via a chat link rather than spoken aloud).

2. Speaker & Participant Disambiguation (Tutor vs. Student Separation)

Current Limitation: Evaluating the full frame's visual telemetry (mean brightness, contrast, background) aggregates the entire meeting canvas. If a student's webcam is dark, poorly lit, or lacks a virtual background, the global metric is polluted, falsely penalizing the teacher.

Proposed Enhancement: Implement a dynamic Region-of-Interest (ROI) segmentation pipeline:

Visual Stream Separation: Use lightweight layout detection (e.g., YOLO or MediaPipe face mesh) to detect multiple video feeds inside the Zoom/Google Meet grid and isolate the Tutor's specific tile (identifiable via fixed coordinates, host badges, or active-speaker borders). Visual telemetry (8×8 background check, lighting, eye contact) should be calculated strictly on the tutor's cropped ROI.

Audio Speaker Diarization: Integrate a diarization model (such as pyannote-audio or Whisper with diarization extensions) to separate the audio track into distinct speaker turns (SPEAKER_00: Tutor, SPEAKER_01: Student). This ensures audio tone and engagement metrics evaluate the teacher's delivery rather than student background noise.

3. Domain-Specific Video Perception Models

Current Limitation: Using zero-shot LLM prompts on sampled static keyframes misses temporal dynamics (e.g., screen-sharing pacing, camera toggling throughout the hour, posture shifts, active board manipulation).

Proposed Enhancement: Move to a dedicated Vision-Language Model (VLM) or domain-fine-tuned classifier:

Fine-tune a specialized vision encoder (e.g., a lightweight CLIP or SigLIP backbone) specifically on chess coaching screenshots to identify:

Presence and correct alignment of the branded 8×8 chess background.
Active board share vs. talking head vs. camera-off states.

Implement an event-based temporal sampler: Instead of sampling frames at fixed intervals (e.g., every 6 minutes), sample keyframes triggered by state transitions (e.g., when the active window changes or when camera states toggle), giving higher temporal fidelity without increasing frame count.

4. Acoustic Signal Processing Over Zero-Shot Prompting

Current Limitation: Relying solely on textual transcript descriptions or zero-shot LLM inferences to grade vocal tone and warmth ignores the acoustic dimensions of voice.

Proposed Enhancement: Implement an end-to-end acoustic evaluation pipeline:

Prosodic & Pitch Analysis: Extract fundamental frequency (F_0), pitch variance, and speech rate using signal processing libraries (such as librosa or praat-parselmouth). Energetic and engaging delivery correlates with dynamic F_0 contour variations, whereas flat, monotonous speech shows minimal pitch divergence.

Speech Emotion Recognition (SER): Run an audio model (such as wav2vec2-emotion or an open SER transformer) directly on extracted teacher audio segments to quantify vocal warmth, positivity, and patience natively from the raw waveform rather than guessing from text tokens.

Dead-Air & Latency Metrics: Measure silence ratios and student response wait times programmatically by calculating Voice Activity Detection (VAD) ratios (using Silero VAD), providing objective, data-backed metrics for "general interactive flow."

3 .Other approaches in my mind that I would have tried

1. Multimodal learning
2Better scoring rather compared to the quick and dirty approach I tried
3. A more agentic approach rather than quick and dirty Multistep LLM pipeline or LLM as a judge approach
4. Better feature capturing across transcripts and across the video or audio
5. Making the model more grounded to the datasets provided - primarily affected to resource limitation I had to go for LLM as judge approach or not even agentic AI apporach where the model can correct things by learning from the data
6. Wish I had more experince in video processing and audio processing